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EXPERIMENT 5

Aim: To Study Architecture of OpenStack cloud computing platform.

Theory:

OpenStack is an open-source cloud computing platform that provides Infrastructure-as-a-Service (IaaS). It consists of several interrelated components that control hardware resources such as compute, storage, and networking. The architecture of OpenStack is modular, allowing different components to work together to manage cloud environments efficiently.

OpenStack Architecture Overview

OpenStack is composed of multiple services that interact through APIs. The major components of OpenStack include:

1. Compute (Nova)

- Manages virtual machines (VMs) and provides an abstraction layer over the hypervisors.
- Supports multiple hypervisors, including KVM, QEMU, VMware, and Hyper-V.
- Handles scheduling, provisioning, and management of compute resources.

2. Networking (Neutron)

- Provides networking as a service, enabling complex network topologies.
- Supports SDN (Software-Defined Networking), VLANs, VXLANs, and firewalls.
- Allows users to create virtual networks, routers, floating IPs, and security groups.

3. Storage

- **Block Storage (Cinder):** Provides persistent storage for VMs similar to traditional hard drives.
- **Object Storage (Swift):** A scalable, redundant storage system for unstructured data.
- **Shared File System (Manila):** Offers shared file storage services similar to NFS.

4. Identity Management (Keystone)

- Handles authentication and authorization across OpenStack services.



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- Supports multi-tenancy, role-based access control (RBAC), and integration with LDAP/AD.

5. Image Service (Glance)

- Stores and manages disk images for VM provisioning.
- Supports various image formats like QCOW2, VMDK, and RAW.

6. Dashboard (Horizon)

- A web-based UI for managing OpenStack services.
- Provides an easy-to-use interface for administrators and users.

7. Orchestration (Heat)

- Automates the deployment of infrastructure using templates (HOT - Heat Orchestration Template).
- Works like AWS CloudFormation.

8. Telemetry (Ceilometer, Aodh, Gnocchi, Panko)

- Monitors and collects performance and usage metrics.
- Enables billing, autoscaling, and resource optimization.

9. Bare Metal Provisioning (Ironic)

- Provides support for provisioning and managing bare metal servers instead of VMs.

10. Containers (Magnum, Zun)

- Manages Kubernetes, Docker, and other container orchestration platforms.

11. DNS Service (Designate)

- Offers DNS as a service, automating domain name management.

12. Messaging Queue (RabbitMQ)

- Enables communication between different OpenStack components.

Logical Architecture

The logical architecture of OpenStack consists of three main layers:

1. **Dashboard (Horizon):** Provides a graphical interface for users to manage resources.
2. **OpenStack Core Services:** Includes compute, networking, storage, identity, and image services.



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3. **Underlying Infrastructure:** Comprises hypervisors, storage backends, and network components.

Deployment Models

OpenStack can be deployed in various ways:

- **Public Cloud:** Hosted by service providers like Rackspace.
- **Private Cloud:** Deployed on an organization's own infrastructure.
- **Hybrid Cloud:** A mix of private and public clouds.

Conclusion

OpenStack is a flexible, scalable cloud platform used for building both private and public cloud environments. Its modular architecture allows customization based on specific cloud computing needs.